

Programming for Business (CL-2016)

LAB INSTRUCTOR
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Week 03 Objectives:

- Data type sizes in memory
- Bitwise Operator
- Operator Precedence
- Type Casting
- Random number
- String with different data bases
- Type Conversion
- Creating Ranges

INT	28
FLOAT	24
Empty String	41/49
Character in String	1
Boolean	28

```
import sys
integer = 10
print("The size of the integer variable is:",sys.getsizeof(integer), "bytes.")
```

Bitwise Operators

- Python Bitwise operators act on bits and perform bit-by-bit operations. These are used to operate on binary numbers.
- Bitwise Operators in Python are as follows:
 - Bitwise NOT
 - Bitwise Shift
 - Bitwise AND
 - Bitwise XOR
 - Bitwise OR

Bitwise operators are used to compare (binary) numbers:

Operator	Name	Description
• &	AND	Sets each bit to 1 if both bits are 1
•	OR	Sets each bit to 1 if one of two bits is 1
• ^	XOR	Sets each bit to 1 if only one of two bits is 1
• ~	NOT	Inverts all the bits
• <<	Zero fill left shift	

Shift left by pushing zeros in from the right and let the leftmost bits fall off

- >> Signed right shift

Shift right by pushing copies of the leftmost bit in from the left, and let the rightmost bits fall off

Example

- `a = 1`
- `b = 4`

`print(a & b)`

`print(a | b)`

`print(~a)`

`print(a ^ b)`

`print(a << 2)`

`print(a >> 2)`

Operator Precedence

Operators	Definitions	Precedence
()	Parenthesis	Highest
**	Exponentiation	
~	Bitwise NOR	
+, -	Sign (positive, negative)	
*, /, //, %	Multiplication, division, floor division, modulus division	
+, -	Addition, subtraction	
&	Bitwise AND	
^	Bitwise XOR	
	Bitwise OR	
<, <=, >, >=, ==, !=, is, is not	Relational operators, membership operators	
not	Boolean (Logical) NOT	
and	Boolean (Logical) AND	
or	Boolean (Logical) OR	
		Lowest

Example

- `print((6 + 3) - (6 + 3))`
- `print(100 + 5 * 3)`

Addition and subtraction has the same precedence, and therefore we evaluate the expression from left to right:

- `print(5 + 4 - 7 + 3)`

Python Type Casting

Integers:

- `x = int(1)`
 `y = int(2.8)`
 `z = int("3")`

Floats:

- `x = float(1)`
- `y = float(2.8)`
- `z = float("3")`
- `w = float("4.2")`

Example

Strings:

- `x = str("s1")`
 `y = str(2)`
 `z = str(3.0)`

Boolean

- `x = bool(1)`
- `y = bool(2.8)`
- `z = bool("")`

Type Conversion

- `x = 1`
`y = 2.8`
`z = 1j`

convert from int to float:
`a = float(x)`

convert from float to int:
`b = int(y)`

convert from int to complex:
`c = complex(x)`

```
print(a)
print(b)
print(c)
```

```
print(type(a))
print(type(b))
print(type(c))
```

Square Root

To find the square root, the input has to be converted into a number:

```
import math
```

```
x = input("Enter a number:")
```

```
y = math.sqrt(float(x))
```

```
print(f"The square root of {x} is {y}")
```

Python program to convert int to exponential

```
x = 3e4  
print(x)
```

```
num = 456789  
print(f"{num:e}")
```

- Up to certain decimal place

```
print(f"{num:.2e}")
```

How to Reverse a String in Python

Example

- Reverse the string "Hello World":
- `txt = "Hello World"[::-1]`
`print(txt)`
- `txt = "Hello World"[::-2]`
`print(txt)`

Converting Strings with Different Bases

Binary string

- `s = "1010"`
- `num = int(s, 2)`
- `print(num)`

Hexadecimal string

- `s = "A"`
- `num = int(s, 16)`
- `print(num)`

Python range

Creating ranges

- The range() function can be called with 1, 2, or 3 arguments, using this syntax:
- range(start, stop, step)

Create a range of numbers from 0 to 9:

- x = range(10)
- x = range(3, 10)
- x = range(3, 10, 2)

Example

Convert different ranges to lists:

- `print(list(range(5)))`
- `print(list(range(1, 6)))`
- `print(list(range(5, 20, 3)))`

Slicing Ranges

- `r = range(10)`
 `print(r[2])`
 `print(r[:3])`

Random Number

- Python does not have a random() function to make a random number, but Python has a built-in module called random that can be used to make random numbers:

Example

- Import the random module, and display a random number between 1 and 9:

```
import random
```

```
Print(random.random())
```

```
print(random.randrange(1,10))
```

```
Print(random.randrange(1,10,2))
```

String Length

- To get the length of a string, use the len() function.
- Example
- The len() function returns the length of a string:
- `a = "Hello, World!"`
- `print(len(a))`

Change Item Value

- To change the value of a specific item, refer to the index number:

Example

- Change the second item:

```
thislist = ["apple", "banana", "cherry"]  
thislist[1] = "blackcurrant"  
print(thislist)
```

Membership Testing

- `r = range(0, 10, 2)`
- `print(6 in r)`
- `print(7 in r)`

Length

- `r = range(0, 10, 2)`
`print(len(r))`

Conversion of a number

- `num = int(input("Enter an integer: "))`
- `print("Binary:", bin(num))`
- `print("Octal:", oct(num))`
- `print("Hexadecimal:", hex(num))`

Convert Binary to Decimal

- `binary = input("Enter a binary number: ")`
- `decimal = int(binary, 2)`
- `print("Decimal value:", decimal)`

Convert Hexadecimal to Decimal

- `hexa = input("Enter a hexadecimal number: ")`
- `decimal = int(hexa, 16)`
- `print("Decimal value:", decimal)`

Final Lab Task

- Swap two numbers without third variable using arithmetic and bitwise operator.
- Take amount from user, Convert transaction amount into float(2 decimals) and scientific notation(exponential up to 3 decimal places) for reporting.
- Convert an account ID to binary and hex for secure storage.
- Show balance increments for savings accounts in multiples of 3(range)
- Take transaction ID as string input , find length and slice it up to any number of your choice.